

Unknot Recognition by Continuation Quotients and Certified Algebraic Splitting

Short interval models and exact hidden summands
A quantitative route to quasi-polynomial time

Research continuation for the ProveIt project

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Abstract

We continue the exact ProveIt unknot recognizer with a decision-specific algebraic quotient and a verified method for splitting hidden summands. For a direct summand whose differential entries are scalar multiples of one fixed square-zero endomorphism, classical type- A decomposition produces interval complexes. We prove that every such interval of length at least two can be replaced by a two-term interval while preserving the completed unreduced Khovanov rank capped at three under every remaining tangle continuation. The proof uses the dual-number identity

$$\dim_{\mathbb{F}_2} H(J_\ell \otimes_A D) = a\ell + 2 \sum_m b_m \min(\ell, m)$$

over $A = \mathbb{F}_2[\epsilon]/(\epsilon^2)$, where J_ℓ is the free ϵ -string, and the even-dimensional circle spaces in an ordinary closure. The length bound is sharp for these observations. Exact homology calculations retain the original interval lengths.

At a stage where all remaining summands admit this normalization, the number of physical objects is at most $P(w) = (w-1)!!(1 + 2(2^{w/2-1} - 1))$, independent of interval lengths and virtual multiplicities. If g bounds the number of crossings between such stages, we obtain

$$T(n; w, g) \leq n^{O(1)} \exp O\left(g + w \log(w+1) + 2^{w/2}\right).$$

This is quasi-polynomial under an explicit condition on w and g ; no universal bound on these execution parameters is established.

The implementation also solves bounded scalar chain-endomorphism systems and verifies Fitting decompositions, exposing genuine direct sums inside connected differential graphs. A separate subset dynamic programme exactly optimizes crossing windows. Tests include independent dense continuation calculations, witness replay, complete homology comparisons, and paired timings. The measurements distinguish fewer algebraic operations from wall time and distinguish synthetic activation of interval shortening from observed behavior on actual diagrams. The deliverable includes the article, source, reference snapshots, integration patches, and reproducible evidence.

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1 Results, scope, and the verified starting point

1.1 The question this continuation answers

The ProveIt project asks for an exact recognizer whose running time approaches

$$T(n) = n^{O(\log n)} = 2^{O((\log n)^2)} \quad (1)$$

on an explicitly given classical knot diagram with n crossings. The existing implementation is complete without resource limits, but its worst-case bound is exponential. This article develops a specific way of discarding algebraic information that is irrelevant to the final decision, and a verified way of exposing direct summands that were hidden by the current choice of basis.

The strongest new statement is an observation theorem. Suppose a genuine direct summand of the open complex has one crossingless matching M in every degree and one fixed nonzero nilpotent coefficient θ in every nonzero differential entry. First put its scalar coefficient matrices into interval form. Then, *for the unreduced rank-two unknot decision*, every interval of length at least two may be replaced by an interval of length two. This preserves the final total rank capped at three under every remaining crossing sequence and closure. The replacement is generally not a chain homotopy equivalence. Its safety follows from a proved property of every allowed continuation, rather than from agreement on a finite test collection.

The same observation gives an explicit complexity theorem. At a stage whose residual components all have the stated form, the number of stored objects after decision compression is bounded solely in terms of the frontier size. If such stages occur frequently enough, the entire scanner has a quasi-polynomial bound. The missing universal assertion is now a concrete structural condition on the intervening residual complexes.

Contribution	Guarantee	Status
Interval normalization	Exact chain isomorphism for common-nilpotent direct summands, with interval counts computed by binary rank invariants	Proved and implemented
Length-two decision model	Same completed rank capped at three for every ordinary Khovanov continuation	Proved and implemented; exact ranks excluded
Scalar Fitting splitting	Every accepted split is an exact chain isomorphism, with complete transformed cross terms checked	Implemented with bounded search
Checkpoint complexity	Explicit $2^{O((\log n)^2)}$ sufficient condition in frontier size and distance between suitable stages	Proved conditional bound
Window order optimization	Exact optimum for a chosen window, with a nonincreasing global combinatorial objective	Proved and implemented
General quasi-polynomial recognizer	Bound (1) on every input	Not established

1.2 Repository and artifact provenance

The inspected ProveIt revision [1] is `eb79136d949a1c0e6f4bd2f07e6b3990d1130632`. Its `Topology/UnknotRecognition/fast` tree has Git identifier `b99a666269f660d932fc82ac8d1f29d3ae42231f`. All forty source, test, example, and configuration files in the preceding package's unchanged reference directory were independently hashed as Git blobs and matched to this current tree. Thus the starting implementation is grounded in repository bytes, rather than a reconstruction from a conversation summary. The package records both Git blob identifiers and SHA-256 hashes.

The immediate development base is the separately delivered *Unknot Recognition by Structural Certificates and Compressed Chain Complexes*, version 0.3.0 [2]. That package adds signed Seifert-graph certificates, literal component sharing, multiplicity saturation, and optional suffix-Euler lower bounds. Those additions are retained in the supplied version 0.4.0. They are not contributions of the present article.

A concurrent continuation investigated faster factorization, sparse Alexander calculations, pointed homology, and a three-braid recognizer. The present package is based on the inspected repository plus the structural 0.3.0 package; it does not claim that the independent branch has been merged or jointly tested. The two supplied patches distinguish an incremental update from 0.3.0 and a cumulative update from the pinned repository tree. This makes the integration boundary explicit.

1.3 Relation to the two linked research repositories

We also inspected the `openai/math` repository at `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Its catalogue does not provide a ready-made replacement for the knot recognizer. One nearby theme is family 116, concerning finite matrix observations of bounded noncommutative formulas. Its formalization scope distinguishes a division-free hitting property from separately asserted construction and dimension bounds [3]. The useful methodological connection is to prove which finite observations suffice before compressing an algebraic object.

No identity-testing theorem is imported as a Khovanov rank algorithm. A small formula, a small chain model, and a method for evaluating homology rank are different inputs. Our continuation theorem explicitly proves the needed observation property in the dual-number setting, where the relevant derived category can be classified. That is the substantive bridge to the broader compressed-algebra theme.

1.4 Current primary literature and the size of the target

Lackenby’s 2021 lecture material announces $2^{O((\log n)^3)} = n^{O((\log n)^2)}$ time [12]. This is quasi-polynomial, but differs from the stronger target (1) used in this project. His July 2026 preprint gives a new hierarchy algorithm and efficient certification results. Section 9 expressly treats an iteration bound separately from an implementation time bound; Proposition 9.1 gives $L(g+1)^L$ iterations in its hierarchy parameters [13]. The sources inspected do not supply an end-to-end quasi-polynomial implementation of the geometric route. This article advances the existing algebraic scanner.

A particularly relevant obstruction is the 2026 work of Kelomäki and Schütz. Their Theorem 1.2 proves exponential behavior of the explicit Bar-Natan scanning and divide-and-conquer algorithms even on positive or alternating three-braids. Their separate special-purpose result computes three-braid homology in polynomial time [11]. Thus a bound on boundary matching types alone cannot control the number of basis objects or the differential data between them. Both multiplicity compression and compression within a single connected component matter.

The classical inputs here are the tangle construction and local cobordism relations [4, 5], type- A interval decomposition [7], and the derived classification over the dual numbers [8]. We give proofs of the algebraic statements actually used. We make no literature-wide priority claim for barcode decomposition, Fitting decomposition, or the classification itself. The contribution is their specific continuation quotient, verified implementation, and quantified application to this recognizer.

1.5 Encoding and validation conventions

The input is an explicitly listed planar-diagram code, or a braid/grid description converted to such a code. Public command-line input is validated as a one-component classical spherical diagram.

The low-level scanner APIs retain the inherited validity promise. An empty diagram denotes one crossing-free circle, with unreduced rank two. Normalized edge labels require $O(n \log(n + 2))$ input bits. Arbitrarily long supplied labels also incur their reading and normalization cost.

All new algebra is over $\mathbb{F}_2$. The scanner forgets quantum degree and retains raw cube homological degree in exact-rank mode. In decision mode, uniform homological shifts and then interval lengths may be discarded only for the proved total-rank observation. The distinction is reflected in separate return schemas: **rank** and **by_degree** are exact, whereas **rank_capped** records only a threshold result. A resource limit produces UNKNOWN, never a mathematical verdict inferred from an unfinished scan.

2 The algebraic model and the decision threshold

The input is an explicitly listed, validated classical knot diagram with n crossings. Complexity statements use normalized edge labels of $O(\log(n + 2))$ bits; reading and normalizing a longer external encoding incurs its actual input-size cost. The empty input denotes one crossing-free circle. A scan order determines, after each crossing, a finite set of live boundary labels. Its cardinality is even. As in the preceding implementation, the objects of the scanning category are loopless matchings of these labels; closed circles are replaced by their two summands by delooping. We retain homological degrees and work over $\mathbb{F}_2$, but forget the quantum grading. In particular, the basis changes below are assertions in the quantum-ungraded category. A quantum-graded extension would require additional compatibility conditions on those basis changes.

Write

$$r_2(K) = \dim_{\mathbb{F}_2} \text{Kh}(K; \mathbb{F}_2).$$

The decision criterion used by the existing recognizer is

$$K \text{ is an unknot} \iff r_2(K) = 2. \tag{2}$$

Here is the coefficient justification. The reduced/unreduced splitting over $\mathbb{F}_2$ gives $r_2(K) = 2 \dim_{\mathbb{F}_2} \widetilde{\text{Kh}}(K; \mathbb{F}_2)$ [10]. The universal coefficient theorem implies

$$\dim_{\mathbb{Q}} \widetilde{\text{Kh}}(K; \mathbb{Q}) \leq \dim_{\mathbb{F}_2} \widetilde{\text{Kh}}(K; \mathbb{F}_2).$$

The rational-rank unknot-detection theorem then excludes every nontrivial knot when the reduced $\mathbb{F}_2$ rank is one [9]. The reverse implication follows from the unknot complex. This argument does not assume that a rank-one reduction modulo two excludes odd integral torsion.

We use the notation

$$\text{cap}_3(a) = \min(3, a), \quad a \in \mathbb{N}.$$

A completed capped value of two therefore certifies UNKNOWN, while a value of three certifies KNOTTED. For a valid knot the latter value means that the actual rank is at least four. Values zero and one indicate a violated input promise or an implementation invariant; they are not interpreted as topological verdicts.

The algebraic correctness statements are relative to the established Bar–Natan gluing, delooping, and cancellation operations [4, 5]. The new algorithms operate on their finite matrices and use exact binary arithmetic. They do not constitute a formal verification of the entire topological foundation.

3 Exact interval normalization of square-zero components

3.1 A recognizable coefficient subcategory

For a matching m with k arcs, the endomorphism ring used by the scanner is the dot algebra

$$R_m = \text{End}(m) \cong \mathbb{F}_2[x_1, \dots, x_k]/(x_1^2, \dots, x_k^2). \quad (3)$$

Its binary basis consists of the squarefree dot monomials; its dimension is 2^k . The identity is the constant monomial. This is the familiar diagonal endomorphism algebra of the matching category [4, 5].

Let $\mathfrak{m} = (x_1, \dots, x_k)$ be the augmentation ideal. Every $\theta \in \mathfrak{m}$ satisfies $\theta^2 = 0$: in characteristic two, the square of a sum is the sum of its squares, and each nonconstant squarefree monomial has square zero. Conversely, an element with constant coefficient one is a unit. Thus the nonzero augmentation elements are exactly the nonzero nonunits of this ring.

The undirected differential graph has one vertex per object and an edge when a differential entry is nonzero. Its connected components are literal direct summands of the complex. The earlier sharing backend identifies repeated components by complete serialization [2]. The following condition exposes additional summands inside such a component.

Definition 3.1 (Pure component). *A nonsingleton differential component is pure if all its objects have one matching m and every nonzero differential entry is one fixed nonzero $\theta \in R_m$ satisfying $\theta^2 = 0$. A singleton is pure without a choice of θ .*

Purity is a test of the actual stored matrix. We do not posit an oracle which determines whether an arbitrary complex can be made pure. In a pure component, write the degree- h term as m^{d_h} . Then

$$\delta_h = \theta M_h, \quad M_h : \mathbb{F}_2^{d_h} \longrightarrow \mathbb{F}_2^{d_{h+1}}, \quad (4)$$

where δ_h is the differential and M_h is a binary matrix. The matrices M_h need not satisfy $M_{h+1}M_h = 0$. The actual differential squares to zero because $\theta^2 = 0$. The distinction is essential for the algorithm: its rank calculations use products of the M_h , not products of the actual cobordism differentials.

Definition 3.2 (Interval complex). *For integers $a \leq b$, let $I_{[a,b]}(m, \theta)$ have one copy of m in each degree $a, \dots, b$, with differential θ between successive copies and zero elsewhere. Its length is $b - a + 1$. After removing the birth shift a , write this complex as $I_{b-a+1}(m, \theta)$.*

3.2 The exact decomposition and its computation

Theorem 3.3 (Exact interval normalization). *Every pure component is chain-isomorphic to a direct sum of interval complexes. The isomorphism is given by binary changes of basis within each homological degree. The interval multiplicities are uniquely determined by the binary matrices in (4).*

Proof. We first prove the elementary interval decomposition for an arbitrary finite sequence of binary vector spaces and maps M_h ; this is the equioriented type- A case of the usual interval classification [7]. Suppose the sequence is nonzero. Let a be its earliest nonzero degree, and choose the largest $b \geq a$ for which the composite from V_a to V_b is nonzero, counting the identity when $b = a$. Choose v_a whose image v_b is nonzero, and let v_h be its successive images for $a \leq h \leq b$. Choose a covector λ_b with $\lambda_b(v_b) = 1$, and define

$$\lambda_h = \lambda_b M_{b-1} \cdots M_h, \quad p_h = v_h \lambda_h \quad (a \leq h \leq b).$$

Outside this interval put $p_h = 0$. We have $\lambda_h(v_h) = 1$, so each p_h is idempotent. For an interior arrow, $p_{h+1}M_h = M_hp_h$. At the left endpoint $V_{a-1} = 0$; at the right endpoint $M_bv_b = 0$ by maximality of b . Hence p is an idempotent endomorphism of the entire sequence. Its image is one interval and its kernel is an invariant complement. Induction on the total vector-space dimension proves existence.

Every binary invertible change-of-basis matrix remains invertible after its entries are multiplied by id_m in the additive category: its binary inverse supplies the actual inverse. Applying the changes of basis to θM_h therefore gives a chain isomorphism of the original component with the corresponding θ -intervals. The multiplicity formula below proves uniqueness. $\square$

Let the occupied degree range be $u, \dots, v$. For $u \leq a \leq b \leq v$ define the rank invariant

$$\rho(a, b) = \text{rank}(M_{b-1} \cdots M_a), \quad \rho(a, a) = d_a. \quad (5)$$

Ranks with an endpoint outside the occupied range are assigned the value zero. An interval $[p, q]$ contributes one to $\rho(a, b)$ exactly when $p \leq a \leq b \leq q$. Taking successive differences gives

$$\mu(a, b) = \rho(a, b) - \rho(a-1, b) - \rho(a, b+1) + \rho(a-1, b+1). \quad (6)$$

This is the multiplicity of $[a, b]$. In particular, the result is a nonnegative integer, and

$$\sum_{a \leq b} \mu(a, b)(b - a + 1) = \sum_h d_h. \quad (7)$$

These identities give both a deterministic algorithm and useful independent consistency checks. The implementation can compute all ranks by binary matrix multiplication and elimination, form the differences, and reconstruct the rank invariant from the resulting intervals.

If there are H degree layers and B objects, a conservative elementary bound is $O(H^2 B^3)$ binary field operations. No efficiency claim depends on this exponent being optimal. Bit-packed matrices provide the practical implementation. Recognition of purity and the required coefficient comparisons are polynomial in the stored matrix and coefficient lengths.

Replacing the component by explicit intervals retains exactly B objects by (7). Its advantage is that these objects now lie in separately shareable summands. Number an interval's vertices in increasing degree, and remove its birth shift. Its complete template is then determined by $(m, \theta, \text{length})$. Length-one templates do not use θ . Equal templates can be continued once, with exact shifted multiplicities or with the existing saturated multiplicities.

Example 3.4 (A strict improvement over graph-component sharing). *Take $B \geq 2$ copies of m in degree zero and B copies in degree one, and put θ in every differential entry. Its differential graph is the connected graph $K_{B,B}$, and no entry is a unit. Literal component sharing leaves one template with $2B$ objects. The scalar matrix is the all-ones matrix of rank one. Interval normalization produces one length-two interval and $B-1$ singletons in each of the two degrees. After shifted sharing, only three objects remain: two in the interval and one singleton with multiplicity polynomial $(B-1)(1+z)$.*

This is a valid algebraic stress family. It is not asserted that arbitrary members occur as intermediate states of the current knot scanner. That additional realizability assertion would need a separate construction.

The replacement must apply to an entire direct summand. A tower attached to other matchings may be normalized only if its basis changes are also applied to every external incident differential. Simply discarding those attachments would invalidate the complex. The exact splitter developed in the next part of the article addresses this issue by producing and checking an actual change of basis.

4 Short intervals with identical capped continuation ranks

Exact normalization still leaves intervals of unbounded length. They need not split further: finite free strings are indecomposable over the dual numbers [8]. The decision problem permits a different operation. A long string can be replaced by a short one when every possible remaining continuation has the same capped homology rank. Such a replacement is not asserted to be a chain-homotopy equivalence.

4.1 The dual-number tensor calculation

Put $A = \mathbb{F}_2[\epsilon]/(\epsilon^2)$, and let J_l be the complex of l copies of the free rank-one A -module, in consecutive degrees, with every differential multiplication by ϵ . All tensor products of complexes below are totalized, and all dimensions refer to total $\mathbb{F}_2$ -homology. Homological shifts do not affect these dimensions.

Lemma 4.1 (Tensor pairing of intervals). *For positive integers l, m ,*

$$\dim_{\mathbb{F}_2} H(J_l \otimes_A J_m) = 2 \min(l, m), \quad \dim_{\mathbb{F}_2} H(J_l \otimes_A \mathbb{F}_2) = l.$$

Proof. For the first formula, identify the lm free basis positions of the total tensor product, after forgetting their grading, with

$$R = \mathbb{F}_2[u, v]/(u^l, v^m).$$

The total differential on $A \otimes_{\mathbb{F}_2} R$ is multiplication by ϵ times the linear operator T given by multiplication by $u + v$ on R . There are no sign distinctions in characteristic two. This differential has binary rank $\text{rank}(T)$, so its total homology has dimension $2lm - 2 \text{rank}(T) = 2 \dim \text{coker}(T)$. But

$$\text{coker}(T) \cong \mathbb{F}_2[u, v]/(u^l, v^m, u + v) \cong \mathbb{F}_2[u]/(u^{\min(l, m)}),$$

which has dimension $\min(l, m)$. Collecting the degreewise matrices into one graded total differential preserves the sum of their ranks. For the second formula, ϵ acts by zero on the simple module, so all differentials vanish and there are l one-dimensional terms. $\square$

The needed derived-category classification is particularly small. Every bounded complex of finite A -modules is isomorphic in the bounded derived category to a finite direct sum of shifts of the J_m and shifts of the simple module $\mathbb{F}_2$ [8, Example 3.7(b)]. Appendix A gives an elementary proof using minimal free resolutions and the finite interval argument above. We use this established classification as an algebraic input; no novelty claim is made for it.

Theorem 4.2 (The continuation rank formula). *Let D be a bounded complex of finite-dimensional A -modules. There are nonnegative integers $a, b_1, b_2, \dots$, only finitely many nonzero, such that, for every $l \geq 1$,*

$$r_l(D) := \dim_{\mathbb{F}_2} H(J_l \otimes_A D) = al + 2 \sum_{m \geq 1} b_m \min(l, m). \quad (8)$$

The integer a counts the simple-module summands of D , and b_m counts its free interval summands of length m , ignoring shifts.

Proof. The complex J_l is bounded and free, hence tensoring with it preserves quasi-isomorphisms and computes the derived tensor product. Decompose D as above and apply Lemma 4.1 to each summand. The decomposition is finite, and homology dimensions add under direct sums. This proves the formula simultaneously for all l . $\square$

The theorem does not require the algorithm to compute the decomposition of the potentially enormous suffix D . It proves an identity for every such suffix, which justifies a local replacement before the suffix is processed.

Corollary 4.3 (Universal length cutoff). *For any positive integer threshold c , any D as above, and any $l \geq 1$,*

$$\min(c, r_l(D)) = \min(c, r_{\min(l, c)}(D)).$$

In particular, length three is a safe cutoff for rank capped at three under arbitrary bounded dual-number continuations.

Proof. Only $l \geq c$ needs consideration. If $a > 0$, both ranks are at least c . If $a = 0$ and some $b_m > 0$ has $m \geq c$, that summand contributes at least $2c$ to both ranks. Otherwise every nonzero b_m has $m < c$, so the right side of (8) has stabilized by $l = c$ and the two ranks are exactly equal. $\square$

The cutoff in this fully general statement cannot be lowered uniformly: for $D = \mathbb{F}_2$, the rank is $r_l(D) = l$. The geometric continuations of the scanner have an extra parity property, which permits a sharper cutoff.

4.2 Why a tangle continuation has the required module structure

Fix a matching m and a square-zero endomorphism θ . Let F mean attach all remaining crossing complexes, close the diagram, and apply the ordinary unreduced Frobenius TQFT. For the proof, use the strict raw complex before optional Gaussian contractions. Planar gluing of complexes is additive, preserves composition, and uses the total tensor differential [5]. Thus

$$D = F(m), \quad \epsilon \cdot v = F(\theta)(v)$$

defines a bounded complex of finite A -modules: $F(\theta)$ is a degree-zero chain map and $F(\theta)^2 = F(\theta^2) = 0$. Inspecting the totalized differential gives

$$F(I_l(m, \theta)) \cong J_l \otimes_A D. \tag{9}$$

This identifies the balanced module action required by Theorem 4.2. It is stronger than merely saying that the continuation is a linear operation. The implementation may perform homotopy-preserving contractions, but the raw functor supplies this strict model for the proof.

Lemma 4.4 (Even continuation rank). *If m admits a nonzero square-zero endomorphism, then $\dim_{\mathbb{F}_2} H(F(m))$ is even for every actual scanner continuation.*

Proof. The empty matching has endomorphism ring $\mathbb{F}_2$, so m has at least one arc. Every complete smoothing of the suffix, glued to m , is a nonempty union of circles. Its unreduced TQFT vector space has dimension 2^c for some $c \geq 1$. Hence the raw total chain dimension of $F(m)$ is even. For any finite complex of binary vector spaces,

$$\dim H(D) = \dim D - 2 \operatorname{rank}(d_D),$$

so its total homology dimension has the same parity. This proves the claim. In particular, the argument does not require a marked-point implementation or an assumption that the A -module complex is free. $\square$

Theorem 4.5 (Geometric length cutoff two). *For every matching m , nonzero square-zero $\theta \in \text{End}(m)$, positive length l , and actual remaining continuation F ,*

$$\text{cap}_3(\dim H(F(I_l(m, \theta)))) = \text{cap}_3\left(\dim H(F(I_{\min(l, 2)}(m, \theta)))\right). \quad (10)$$

Proof. By Lemma 4.4 and (8), $r_1(D) = a + 2 \sum_m b_m$ is even, so a is even. Suppose $l \geq 2$. If $a > 0$, then $a \geq 2$ and its contribution at length two is already at least four. If $a = 0$ but some $b_m > 0$ has $m \geq 2$, that summand also contributes at least four at length two. In the remaining case, only b_1 can be nonzero, and $r_l(D) = 2b_1$ is independent of l . These three cases prove the equality. Length one is unchanged. $\square$

Example 4.6 (The geometric cutoff is sharp). *Let m pair four cyclically ordered endpoints as $(1, 2)$ and $(3, 4)$, and close it with the other crossingless pairing $(1, 4)$ and $(2, 3)$. The closure is one circle. Write x_1, x_2 for dots on the two arcs of m and set $\theta = x_1 + x_2$. Under closure both dots act by the same circle operator x , so $F(\theta) = x + x = 0$. The image of m is the two-dimensional circle space. Consequently*

$$\dim H(F(I_l(m, \theta))) = 2l.$$

Replacing a length-two interval by a singleton would change a capped value of three to two. Thus cutoff one cannot preserve these geometric observations uniformly. This example concerns valid algebraic objects and a crossingless planar closure; it does not claim that each source interval is itself the full complex of a knot diagram.

4.3 Compositional correctness of the decision backend

Definition 4.7 (Continuation equivalence). *For two complexes C, C' on the same boundary, write $C \equiv_3 C'$ if every permitted remaining crossing sequence and ordinary closure F satisfies*

$$\text{cap}_3(\dim H(F(C))) = \text{cap}_3(\dim H(F(C'))).$$

This is an equivalence relation by equality of the specified observations. It is preserved by adjoining a common direct summand, taking finitely many copies, and performing a common crossing extension: in the last case, any further completion is also a permitted completion before that extension. Theorem 4.5 identifies long intervals with length-two representatives for this relation. It does not identify their homotopy types.

The prior backend stores a template together with nonnegative multiplicities, optionally recording absolute shifts in a polynomial $p(z) \in \mathbb{N}[z]$ [2]. For total-rank decisions, replace the weight by $\text{cap}_3(p(1))$. Addition and multiplication use saturation:

$$a \oplus b = \min(3, a + b), \quad a \otimes b = \min(3, ab).$$

The map $\mathbb{N} \rightarrow \{0, 1, 2, 3\}$ given by saturation preserves both operations. These are nonnegative counts, not coefficients reduced modulo two.

Theorem 4.8 (Decision correctness under repeated replacements). *Start with the existing exact scanning complex. At any stages, perform exact unit cancellations, exact changes of basis, direct-summand decompositions, and sharing of equal normalized templates. For each pure summand, apply Theorem 3.3; in decision mode shorten each resulting interval to length $\min(l, 2)$ and saturate its multiplicity at three. Assume that structural choices do not depend on the discarded multiplicities or absolute shifts. At final closure the reported count is precisely $\text{cap}_3(r_2(K))$.*

Proof. Fix the full future crossing order at any proposed replacement. By Theorem 4.5, completing a single replaced interval would give the same capped total homology rank. Completion is additive on independent summands and commutes with homological shifts. Since capping commutes with addition and multiplication of nonnegative integers, replacing any number of independent copies preserves the capped rank of the completed whole complex. Exact changes of basis, contractions, and component sharing preserve the actual homotopy type before capping and hence preserve this observation as well.

Repeat this argument at every later replacement, comparing the counterfactual full completion immediately before and immediately after it. Transitivity shows that all replacements together preserve the capped final rank. At closure, the empty matching has endomorphism ring $\mathbb{F}_2$; complete elimination leaves isolated generators, whose capped multiplicities compute that rank. Finally apply (2) for the verdict. $\square$

This proof deliberately uses an observational invariant after shortening. The represented complex need not retain the original homotopy type, exact rank, or degree-by-degree homology. The exact-rank backend therefore uses interval splitting without shortening. The decision backend reports a capped rank and does not supply exact homological counts or cycle representatives. Likewise, exact signed Euler data from the former representation cannot silently be carried through an interval shortening.

The current length alone does not give a completed rank bound greater than two. For a geometric example, take one arc m with $\theta = x$ and close it to one circle. Then $D = A$ with ϵ acting by x , and Lemma 4.1 gives $\dim H(J_l \otimes_A A) = 2$ for every l . Thus arbitrarily long legal interval complexes can have a rank-two closure observation. For arbitrary module continuations, an acyclic D can even give rank zero. The theorem says that the short representative has the same completed threshold observation; it does not license a verdict from the current number of stored objects.

5 A checkpoint bound independent of component size

5.1 The state count after normalization

Fix the implemented ordering and exact splitting policy. Let w be the largest frontier size over the entire planned scan. Let a *pure checkpoint* be a stage at which, after unit elimination and the chosen exact splitting operations, every differential component satisfies Definition 3.1. Include the initial and final states among the checkpoints. Let g bound the number of crossings between successive checkpoints. A checkpoint is recognizable from its current matrix. A useful uniform bound on their gaps is an additional theorem about an input family, not something implied by their definition.

For even $w \geq 0$ put

$$M(0) = 1, \quad M(w) = (w-1)!! \quad (w > 0), \quad N(w) = 2^{2^{w/2}-1} - 1. \quad (11)$$

There are at most $M(w)$ frontier matchings. The quantity $N(w)$ counts the nonzero augmentation elements of the endomorphism ring of a matching with $w/2$ arcs: its $2^{w/2} - 1$ nonconstant basis monomials have arbitrary binary coefficients. In particular, $N(0) = 0$. Counting all perfect matchings avoids assuming that the frontier bounds one disc or that numeric labels specify a useful cyclic order.

Proposition 5.1 (Physical size at a pure checkpoint). *After interval normalization, length cutoff two, and complete sharing, a pure checkpoint satisfies*

$$\begin{aligned} \text{stored templates} &\leq M(w)(1 + N(w)), \\ \text{stored objects} &\leq P(w) := M(w)(1 + 2N(w)), \end{aligned} \tag{12}$$

independently of multiplicities and of the sizes of the original connected differential components.

Proof. For each matching there is one possible singleton type, with one object. Every other retained interval has length two and is determined by its matching and one of the at most $N(w)$ possible nonzero square-zero endomorphisms. It contributes two objects. Constructing the vertices in degree order makes equal triples have equal complete serializations; no graph-isomorphism oracle is needed for these types. Summing these counts gives the bounds. A smaller actual frontier only decreases both M and N . $\square$

For comparison, the exact-rank version retains lengths up to $n + 1$, because every raw cube degree lies between zero and n . Its analogous object bound is

$$P_{\text{exact}}(n, w) \leq M(w) \left(1 + N(w) \frac{n(n+3)}{2} \right), \tag{13}$$

using $\sum_{l=2}^{n+1} l = n(n+3)/2$. Absolute birth shifts are recorded in the exact multiplicity polynomials. The universal cutoff-three variant for arbitrary module continuations, if desired, has the decision bound $M(w)(1 + 5N(w))$ instead of (12).

5.2 Complexity through impure stretches

Theorem 5.2 (Quantified checkpoint complexity). *Under the preceding assumptions, both the exact interval-sharing backend and the shortened decision backend admit the conservative bit-complexity bound*

$$T(n; w, g) \leq n^{O(1)} \exp \left(O \left(g + w \log(w+1) + 2^{w/2} \right) \right). \tag{14}$$

The same expression is a conservative space bound. Exact splitting operations that preserve the number of objects and cost polynomial time in the physical matrix size and coefficient dimension may be included without changing this form of bound.

Proof. In one crossing extension, each old loopless matching has two smoothings. Each smoothing creates at most two new circles, since every new circle contains one of the two newly added smoothing arcs. Delooping therefore creates at most four objects per smoothing, or eight objects per old object in total. Elimination removes objects; exact basis changes and interval normalization do not increase their total number; sharing and shortening only decrease it. Thus, between two checkpoints, the physical object count is bounded by 8^g times the preceding checkpoint bound, allowing an immaterial additional factor eight depending on endpoint accounting.

Unit elimination uses polynomially many matrix and coefficient operations in this physical size. The interval algorithm has polynomial cost by (5)–(6). The same is true of the exact splitting procedures admitted in the statement. In the dot-mask algebra, morphisms have $2^{O(w)}$ coefficient bits, and direct enumeration of monomials and their gluings gives a conservative $2^{O(w)}$ bit cost per composition or transfer, including the four temporary ports of the next crossing. This is the same finite algebra cost model used in the prior analysis [2].

Exact multiplicity polynomials have at most $n + 1$ degree positions. The virtual unreduced allocation grows by at most eight per crossing, so each coefficient has $O(n)$ bits. In decision mode the

weights have constant size. Complete key comparisons, matching identifiers, and persistent caches add polynomial factors in the stated physical size and n ; the bound need not rely on average-case hashing.

Finally,

$$\log M(w) = O(w \log(w + 1)), \quad \log(1 + N(w)) = O(2^{w/2}).$$

Substitute either (12) or (13) into the physical-size estimate, charge the polynomial operations across n crossings, and absorb the resulting fixed powers in the exponential. This gives (14). $\square$

Corollary 5.3 (A specific quasi-polynomial regime). *Suppose a diagram family has constructible scan orders and a proved checkpoint schedule for which*

$$g + 2^{w/2} + w \log(w + 1) = O(\log^2(n + 2)). \quad (15)$$

If constructing the order and schedule costs at most $n^{O(\log n)}$ bit operations, the resulting complete recognizer runs in $n^{O(\log n)}$ bit operations on that family. Fixed w and fixed g give a polynomial bound when the order is polynomial-time constructible.

For integer w , the last term in (15) can be absorbed into a universal multiple of $2^{w/2}$; it is displayed to retain the actual matching-count contribution. For example, $w \leq 4 \log_2 \log(n + 2) + O(1)$ and $g = O(\log^2(n + 2))$ suffice.

The previous component-sharing theorem required a bound on the maximum post-elimination connected component size b , with the sufficient condition $b^2 2^{w/2} = O(\log^2 n)$ [2]. The new result removes that component-size requirement at pure checkpoints. In particular, an interval can have length proportional to n , and a connected pure component can contain arbitrarily many copies mixed by binary changes of basis. Neither prevents the checkpoint bound. The exact and decision variants achieve this in different ways: exact normalization keeps polynomially many possible lengths, while the continuation quotient uses only two lengths.

This remains a conditional structural theorem. No argument here shows that every knot diagram, or every prime nonhomogeneous diagram left after the structural front end, has frequent pure checkpoints. Taking only the initial and final checkpoints can leave $g = n$, which yields an exponential bound. Likewise, a large frontier remains costly because $N(w)$ is doubly exponential in w . Experimental success of an exact splitter on individual diagrams is useful evidence about these parameters, but does not replace the missing uniform theorem.

5.3 A sharper bound for restricted coefficient alphabets

The all-polynomials count $N(w)$ is deliberately conservative. It can be replaced when the coefficients occurring at the useful checkpoints have additional structure. The restriction need only hold at those checkpoints; arbitrary intermediate coefficients are still charged at their full dot-mask size.

Corollary 5.4 (Checkpoint coefficient alphabets). *Suppose that, at every pure checkpoint and for each actual matching m , all nonsingleton pure components use coefficients from a set $\Theta_{i,m}$ with*

$$|\Theta_{i,m}| \leq A(w).$$

The sets may depend on the checkpoint i and matching m ; only their cardinality bound must be uniform. Under the other assumptions of Theorem 5.2, the decision checkpoint has at most $M(w)(1 + 2A(w))$ objects, and both the exact and decision variants admit the bound

$$T(n; w, g, A) \leq n^{O(1)} \exp(O(g + w \log(w + 1) + \log(1 + A(w)))) . \quad (16)$$

Proof. For each matching, retain one singleton type and at most $A(w)$ length-two types, giving the stated object count. The exact variant again gains only a polynomial factor in n from possible lengths. Between checkpoints the same factor 8^g bounds expansion. Importantly, we still charge $2^{O(w)}$ bit operations for general coefficient compositions and transfers; we do not assume that low-complexity coefficients remain so between checkpoints. Its logarithmic contribution $O(w)$ is absorbed by $O(w \log(w+1))$ for positive even w , and the empty-frontier case is constant. Substitution in the preceding proof gives (16). $\square$

For example, put $p = w/2$. If every checkpoint coefficient is a nonzero *linear* combination of the p dot generators, then

$$A(w) \leq 2^p - 1, \quad T \leq n^{O(1)} \exp O(g + w \log(w+1)).$$

More generally, if every such coefficient has dot degree at most a fixed positive integer d , the squarefree monomial basis gives

$$A(w) \leq 2^{S_d(p)} - 1, \quad S_d(p) = \sum_{j=1}^{\min(d,p)} \binom{p}{j}. \quad (17)$$

For fixed d , this yields

$$T \leq n^{O(1)} \exp O_d(g + w \log(w+1) + w^d).$$

In particular, either the linear or the degree-at-most-two hypothesis, together with $w = O(\log(n+2))$ and $g = O(\log^2(n+2))$, suffices for the target $n^{O(\log n)}$. The full coefficient arithmetic is still polynomial in n in this width regime.

These are additional, checkable checkpoint hypotheses. They are not asserted invariants of crossing extension, elimination, or scalar splitting, and the measured corpus does not establish them for a uniform class of residual knot diagrams. Their value is to exhibit a less restrictive route than counting all $2^{2^{w/2}-1}$ augmentation polynomials: a proof controlling the coefficients at recurring checkpoints could permit logarithmic frontier width.

6 Certified scalar splitting of hidden summands

The interval condition is deliberately restrictive. A visible string of equal-matching objects may have additional maps to other objects, and those attachments cannot be dropped. On the initial named diagram sample, the whole-component interval test found no nontrivial eligible component. We therefore add an exact basis-change procedure that can expose a direct sum inside a connected differential graph. Its correctness is unconditional; its bounded search is not complete.

6.1 A finite linear endomorphism problem

Let C be one connected component with B basis objects. Partition them by the pair (homological degree, matching). A scalar map Q may mix objects inside each part, with coefficients in $\mathbb{F}_2$, but may not mix different parts. Writing Q_h for its degree- h blocks, the chain-endomorphism equations are

$$Q_{h+1}d_h = d_h Q_h. \quad (18)$$

The cobordism entries have finite bit-packed bases. Equating each coefficient in (18) produces a homogeneous binary linear system. If the block sizes are $r_1, \dots, r_t$, there are $v = \sum_i r_i^2 \leq B^2$ unknown scalar entries. Zero and the identity are always solutions.

This restriction to scalar maps is useful. It makes the search an ordinary binary linear-algebra problem while still allowing nontrivial changes of basis. It is also a limitation: a decomposition requiring more general cobordism-valued endomorphisms can be missed.

Lemma 6.1 (A cheap exact skip). *If every (degree, matching) block contains one object and the differential graph is connected, every scalar chain endomorphism is a scalar multiple of the identity.*

Proof. Each object has one scalar $q_v \in \mathbb{F}_2$. A nonzero entry $f : v \rightarrow w$ in the differential gives $(q_w - q_v)f = 0$. Since $f \neq 0$ and the scalar field has two elements, $q_w = q_v$. Connectivity propagates this equality to all objects. $\square$

This lemma avoids solving a linear system in a common case. It does not certify indecomposability with respect to every allowed categorical map.

6.2 From an endomorphism to an actual split

For a computed solution Q , choose a power e at least the largest scalar block dimension. The implementation uses the next power of two and repeated squaring. In each block, form

$$U = \ker Q^e, \quad V = \operatorname{im} Q^e.$$

These are ordinary binary vector spaces. Assemble their bases separately in each (degree, matching) block.

Theorem 6.2 (Certified scalar Fitting split). *The spaces U and V form complementary subcomplexes of C . If their total dimensions are both nonzero, changing to these bases gives a nontrivial direct-sum decomposition of C by an exact chain isomorphism in the implemented cobordism category.*

Proof. For an endomorphism of an r -dimensional vector space, the increasing sequence of kernels and the decreasing sequence of images stabilize by exponent r . At a stabilizing exponent, $\ker Q^e \cap \operatorname{im} Q^e = 0$: if $z = Q^e y$ and $Q^e z = 0$, then $y \in \ker Q^{2e} = \ker Q^e$, so $z = 0$. Rank-nullity gives their direct sum equal to the original space.

Equation (18) also holds for Q^e . Thus the differential maps a kernel vector to a kernel vector and an image vector to an image vector, across every pair of matching blocks. Scalar changes of basis inside one matching are invertible matrices of identity cobordisms, so they are genuine isomorphisms of the additive category. Conjugating the whole differential therefore exposes the two subcomplexes. No map incident with the component is omitted: the component was a complete direct summand before the operation. $\square$

The original individual candidate Q need not be invertible, and is not used as the change of basis. The change-of-basis matrices P_h have columns given by the kernel basis followed by the image basis. The transformed maps are

$$d'_h = P_{h+1}^{-1} d_h P_h. \tag{19}$$

Every coefficient of every map crossing between the two selected sides must vanish. This is checked explicitly. Confusing Q with P , transforming only one end of a differential, or ignoring an external attachment would each make the replacement unsound.

6.3 What is searched and what is verified

The default pass considers components with at most 48 objects and at most 1,024 scalar variables. It inspects up to 64 nullspace-basis vectors and 16 deterministically generated linear combinations, with fixed seed 7243. It accepts at most 16 splits in one stage. These are engineering limits, not theorems that larger components are indecomposable. An unexamined component is retained in full.

For every accepted split, the implementation independently checks the coefficientwise commutator, the inverse of each basis matrix, and every cross-block coefficient after (19). Optional witnesses record the original matching and degree labels, the original and transformed differentials, Q , P , the kernel dimensions, and the exponent used. Independent tests reconstruct the conjugation using separate dense binary matrix operations and reject deliberately corrupted witnesses.

Positive and negative search results can be cached. A key records complete scalar-quiver data: matching names are consistently replaced by first-occurrence indices, a uniform degree shift is removed, and every differential coefficient and target index is retained. This preserves the linear equation system. A cached positive result is still checked by recomputing commutation and the transformed cross terms. Cache equality is equality of full data; a hash collision alone cannot identify two complexes.

After a split, graph components are recomputed, and the interval and literal sharing passes act on the exposed summands. Multiplicity ancestry is carried through every change: a summand arising from one weighted parent inherits that parent's weight. Basis transformations are never allowed to mix independent weighted parents without an explicit expansion of those weights.

6.4 Complexity and the present practical limit

With R an upper bound on the number of coefficient coordinates of a morphism, the full scalar equations admit a conservative polynomial bound in B and R ; ordinary dense binary elimination, for example, gives a bound of order RB^6 elementary field operations. The bounded implementation is much smaller on the tested components, but this overhead is still material. The checkpoint theorem remains valid with this pass: scalar splitting does not increase object count, and its cost is polynomial in the stored complex and coefficient lengths.

A successful split can reduce later differential entries and compositions without reducing the number of stored objects immediately. It may or may not expose repeated templates or common- θ intervals. Section 9 reports all of these quantities separately. In particular, finding hidden summands in actual knot scans does not by itself demonstrate that the length-two quotient has been exercised on those scans.

7 Exact optimization of a bounded crossing window

The previous scan-order planner is a fast greedy search with several starting crossings and tie rules. We add a deterministic local optimization with a precise objective. It is independent of the algebraic quotient and remains optional, since a better frontier profile need not yield a faster homology calculation.

7.1 The subset state is exact

Let P be the fixed set of crossings before a window, and let B be the k crossings inside it. For $S \subseteq B$, let $\partial(P \cup S)$ be the set of edge labels occurring an odd number of times among crossings already processed. Write

$$b(S) = |\partial(P \cup S)|.$$

Every edge label occurs exactly twice in the full diagram. Thus this set is exactly the scan frontier, including the cancellation of labels that occur twice at a self-loop crossing. It depends on the subset S , not the order used to reach S .

Only the at most $4k$ labels incident with the window can change. Represent those labels by bit positions, let z encode the initial local frontier, and let t_i be the parity mask of crossing i . If c counts frontier labels outside this local set, then

$$b(S) = c + \text{popcount} \left(z \text{ xor } \bigoplus_{i \in S} t_i \right). \quad (20)$$

All subset widths can be computed using the least significant set bit of each subset. The sets at both ends of the window are fixed, so every frontier outside the window is unchanged by its permutation.

7.2 Why the optimization has two passes

The selected objective is lexicographic: first minimize the global maximum frontier size, then an additive cost. The implementation offers either $c_1(b) = b$ or $c_2(b) = 2^{b/2}$. Frontier sizes are even for a degree-four diagram graph. The latter cost is motivated by the dimension of a dot endomorphism space, and is explicitly only a proxy for full scanner work.

A tempting one-label recurrence for pairs (peak, sum) is incorrect. A path with a smaller early peak but a larger sum can lose to a different path when a later frontier raises both peaks to the same value. Lexicographic comparison is not preserved by the operation that takes a maximum in the first coordinate. Our implementation first solves the bottleneck problem and then the constrained additive problem.

Let W_{out} be the maximum frontier size outside the window. Set $p(\emptyset) = W_{\text{out}}$ and

$$p(S) = \min_{i \in S} \max\{p(S \setminus \{i\}), b(S)\}.$$

The optimal global peak is $W_* = p(B)$. Now set $d(\emptyset) = 0$, forbid every nonempty state with $b(S) > W_*$, and compute

$$d(S) = c_j(b(S)) + \min_{i \in S} d(S \setminus \{i\})$$

over feasible predecessors. Backtracking gives an optimal window order. The original order is feasible, and equal-score replacements are discarded to avoid unnecessary changes.

Theorem 7.1 (Exact window optimization). *For fixed crossings outside a window of size k , the two-pass algorithm finds a permutation minimizing the stated global lexicographic objective. It uses $O(k2^k)$ subset transitions and $O(2^k)$ table entries, in addition to polynomial work for boundary masks and exact integer costs. Every accepted replacement weakly decreases the global peak and strictly improves the selected global pair.*

Proof. Equation (20) gives the exact cost at every state. Every window permutation corresponds to exactly one maximal path through the subset lattice. Conversely, each such path adds one distinct crossing at each step and yields a permutation. The first recurrence is the usual bottleneck recurrence on this acyclic graph, with all fixed outside peaks included in its initial value. After W_* is known, the second recurrence is an ordinary additive shortest path in the induced feasible subgraph. Outside frontiers are fixed, so minimizing the local additive contribution minimizes the global one. The lattice contains 2^k states and $k2^{k-1}$ directed inclusion edges. Integer masks have $O(k)$ local bits; the additive costs have at most polynomially many bits in the input size. Thus the corresponding bit complexity is $2^k \text{ poly}(n, k)$. $\square$

The bounded number of overlapping passes is part of the algorithm. When a time budget expires, the planner returns the last completely verified order. This is still a valid input to the recognizer. If $k = O(\log n)$, a polynomial number of windows costs polynomial time; if $k = O((\log n)^2)$, their cost is quasi-polynomial. These statements concern the planner. They do not establish a width bound for its output, a globally optimal unrestricted order, or a quasi-polynomial homology computation.

The implementation is independently checked against all permutations of many small degree-four multigraphs, including loops and parallel edges. The graph theorem does not require those test graphs to be realizable as knot diagrams. Separate tests on validated knot diagrams verify that the resulting orders preserve homology.

8 Implementation and integration

8.1 The order of operations

Version 0.4.0 retains the preceding front-end structural certificates and the existing simplification and invariant filters. Its inherited scanning architecture follows the established local-complex approach [6]. The standard scanner remains the default. The new optional decision backend applies the following operations after each crossing:

1. Form the exact crossing extension and use the inherited Gaussian elimination to cancel unit maps.
2. For the Fitting backend, inspect bounded connected components, verify candidate scalar splits, and transform the complete differential.
3. Recognize whole common- θ components and compute their interval multiplicities from scalar product ranks.
4. In decision mode, shorten intervals to length at most two; in exact mode, retain their lengths and absolute degree shifts.
5. Share identical interval or residual-component serializations and update their nonnegative multiplicities. Decision multiplicities are capped at three.

The stage log records whether every surviving component was a singleton or was actually recognized and normalized by the interval pass. A skipped nontrivial component does not count as a certified checkpoint. The log also records frontier size, unclassified object counts, and crossing distances between checkpoints.

8.2 Public interfaces

The command line obtains a validated `Diagram` through the inherited loader. The new complete decision routes are

```
python -m fastunknot recognize examples/conway.json --backend barcode
python -m fastunknot recognize examples/conway.json --backend fitting
```

The usual inexpensive front-end checks may decide these examples before the selected backend is needed. The reproduction scripts explicitly request raw backend calculations when measuring scanner work.

For exact homology ranks, the separate routes are

```
python -m fastunknot khovanov examples/conway.json --barcode
python -m fastunknot khovanov examples/conway.json --fitting
```

These retain complete interval lengths. The Python APIs are `barcode_khovanov_rank`, `barcode_khovanov_decide`, `fitting_khovanov_rank`, and `fitting_khovanov_decide`, in the corresponding modules. The Fitting exact-rank entry point rejects nontrivial decision caps. Its optional `record_witnesses=True` argument includes split certificates in the result.

The optional ordering pass is selected, for example, by

```
python -m fastunknot recognize examples/conway.json \
  --backend fitting --order-window 10 --order-passes 2
```

It optimizes overlapping windows using the frontier-mass objective. The planner’s time is included in the recognition budget. Standalone calls to `improve_scan_order` expose both supported objectives and retain a complete valid order if their optimization budget expires.

8.3 Evidence and failure semantics

An exact result contains `rank`, `reduced_rank`, and `by_degree`. A default decision result contains `rank_capped`, `rank_cap=3`, and `length_cap=2`. The latter does not report the model’s ordinary rank as the rank of the original knot. The Python decision APIs also permit retaining full interval lengths or using the universally safe length-three cutoff; these choices still return capped rank evidence rather than an exact-rank schema.

Similarly, after length shortening, a weighted object count measures the decision model, not an isomorphic expansion of the original complex. The implementation changes the statistic name to `weighted_decision_model_objects` and sets a `decision_equivalence_only` flag. This avoids incorrectly interpreting an altered model as a lower bound on the original chain dimension.

The `max_objects` ceiling is checked before constructing a crossing extension, so a normalization that would shrink an oversized temporary model is not performed if the ceiling would first be exceeded. Time budgets are cooperative. The low-level APIs raise `ScanLimit` on exhaustion; the recognition pipeline returns `UNKNOWN`. Neither a large model nor a failed decomposition search is treated as evidence of knottedness.

8.4 Validation boundaries

Checking $d^2 = 0$ is valuable, but it cannot establish observational equivalence of two different complexes. The correctness of length shortening therefore rests on Theorem 4.5, while the tests independently exercise its finite-matrix consequences. The optional `check_d_squared` mode checks that subsequent algebraic operations still produce a complex.

The new normalization mixes quantum degrees because the inherited scanner has already forgotten them. It preserves homological degree in exact mode; its scalar Fitting blocks are explicitly restricted to matching and homological degree. A future bigraded implementation must impose compatible quantum shifts as an additional condition.

8.5 Repository integration

The package contains a complete runnable `fast/` tree, a reference copy of the pinned repository files, and the preceding 0.3.0 reference used for incremental comparison. One patch applies directly to the pinned `Topology/UnknotRecognition/fast` tree; the other applies after the structural-compression

package. Both are generated from complete file bytes and independently applied to fresh reference copies during packaging.

No remote repository modification is part of this deliverable. Integration into a tree that also contains the independent three-braid continuation must resolve shared front-end files and rerun the combined tests. The new algebra modules and ordering module are otherwise self-contained, use only the Python standard library, and retain the project’s MIT-0 license.

9 Correctness evidence and measured performance

9.1 Protocol and comparison boundaries

The main experiment contains nineteen validated knot diagrams: ten named examples, eight unselected seeded random braid closures, and one separately labeled random closure selected during exploration because it admitted a scalar split. The named cases are the eight-crossing hard unknot, Conway, Kinoshita–Terasaka, connected sums of two, three, and eight Conway knots, a 36-crossing five-strand stress braid, and the standard closures $T(3, 11)$, $T(4, 9)$, and $T(5, 8)$. The first eight valid random closures generated with seed 54287 form the unselected set. The selected example is the twenty-fourth valid closure from the same stream; it is excluded from any frequency claim about that set. Every actual input PD code is saved in the raw JSON.

Five backends receive the same initial crossing order: the inherited standard exact scanner; the preceding component-sharing exact scanner; the new exact Fitting scanner; its capped decision variant; and that decision variant with window optimization. Initial order selection uses the best greedy order among at most twelve starting crossings. Its measured cost is recorded separately and is not included in Table 2; the optional DP’s cost is included in its column and deducted from the remaining scan budget. All front-end simplification, polynomial filters, Seifert certificates, and visible-sum factorization are bypassed. These are raw scanner comparisons, not complete recognition-pipeline timings.

There are three interleaved repetitions per input and backend, with rotated backend order. Each call has a three-second cooperative budget and a ceiling of 50,000 physical objects. The Fitting settings are the defaults described in Section 6; DP uses window ten, two passes, and the frontier-mass objective. The environment is Python 3.12.14 on x86-64 Linux 6.18.44 with glibc 2.39. Full platform, source-hash, order, stage-history, result, and time records are included. No other research timing workload ran concurrently, although the shared host and the short samples still introduce ordinary timing variation.

The 285 calls produced 279 completed results and six object-ceiling exits. The exits are the three standard-scanner repetitions on each of the three-fold and eight-fold Conway sums. Every completed exact result agrees with the other completed exact backends, and every capped decision agrees with the corresponding exact rank capped at three. The previous component backend already completes both large sums. Their improvement over the standard explicit scanner is inherited and is not credited to this work.

9.2 Correctness checks that are independent of timings

All 94 integrated unit tests pass after the final implementation changes. The suite exercises binary rank calculations, interval multiplicity recovery, exact graded comparisons on diagrams, scalar changes of basis, witness mutation rejection, cap restrictions, CLI routing, budget outcomes, and exhaustive small crossing-window optima. Existing tests for the inherited recognizer and structural front end remain included.

A separate program, `verification/verify_continuation.py`, imports no recognizer code. It enumerates 861 small bounded dual-number complexes, including nonfree modules, constructs the total tensor differential as an ordinary binary matrix, verifies its square, and computes homology ranks for interval lengths one through six. There are 10,206 threshold comparisons; all pass. The tested dimension patterns are

$$(1), (2), (2, 1), (1, 2), (2, 2), (2, 2, 2).$$

The same program records the rank sequences 1, 2, 3, 4, 5, 6 for the simple one-dimensional continuation and 2, 4, 6, 8, 10, 12 for two copies of it. These expose, respectively, an invalid universal length-two rule without parity and an invalid even-context length-one rule. This finite verification checks the theorem’s consequences; the proof still supplies its universal quantifier.

The archive also contains two actual Conway split witnesses and a dense verifier separate from the scalar solver. It verifies matching and degree preservation, the commutator equation, invertibility of the basis changes, reconstruction of every coefficient in every incident differential, and zero cross-partition entries. A complete replay of the Conway scan reproduces both witnesses and the exact rank. Independent adversarial review additionally compared 84 small nullspaces, 30 dense Fitting conjugations, and 40 exhaustive window-order optima; the review record describes their scope. These are session-level checks, not a claim of external peer review or formal proof of the Python implementation.

9.3 The algebraic splitting is active on actual diagrams

Table 1 isolates deterministic work counts at the same order, comparing the preceding component scanner C_s with the new exact Fitting scanner F . “Entries” is the scanner’s accumulated count of differential transfer entries; it is neither peak memory nor final matrix size. “Compositions” counts the recorded morphism compositions. “Largest component” is the largest stored connected differential component recorded by the corresponding normalization path. The raw stage records allow these counters to be inspected in context.

Case	Entries		Compositions		Largest component		Splits
	C_s	F	C_s	F	C_s	F	
C	1519	1349	86	75	41	38	2
KT	1536	1486	113	113	41	39	2
$C^{\#2}$	6592	6091	283	224	240	160	3
$C^{\#3}$	11967	10707	470	353	240	160	4
$C^{\#8}$	37906	34093	1411	1022	240	160	9
$B_{5,36}$	260339	260339	25106	25106	4696	4696	0
$T(4, 9)$	11713	11685	2301	2301	104	104	1
R_{24}	2999	2740	355	341	77	77	4

Table 1: Selected deterministic operation counts. The selected random input R_{24} is explicitly separate from the eight unselected random cases.

The eight-fold Conway sum gives a clear structural improvement: 1,411 compositions become 1,022, a reduction of approximately 27.6%, and the largest recorded component drops from 240 to 160 objects. There are nine accepted Fitting splits. The transfer-entry count falls from 37,906 to 34,093, about 10.1%. For a single Conway knot, two accepted splits reduce the corresponding counts from 1,519 to 1,349 entries and from 86 to 75 compositions. These are genuine algebraic decompositions inside previously connected graphs, with all surrounding maps included.

None of the eight unselected random cases admits an accepted scalar split under the fixed search bounds. The selected random case admits four, but its selection prevents using it to estimate prevalence. On the stress braid, all recorded entries, compositions, and the largest component remain unchanged. This is consistent with an exact but incomplete bounded search; it is not evidence that no other algebraic decomposition exists.

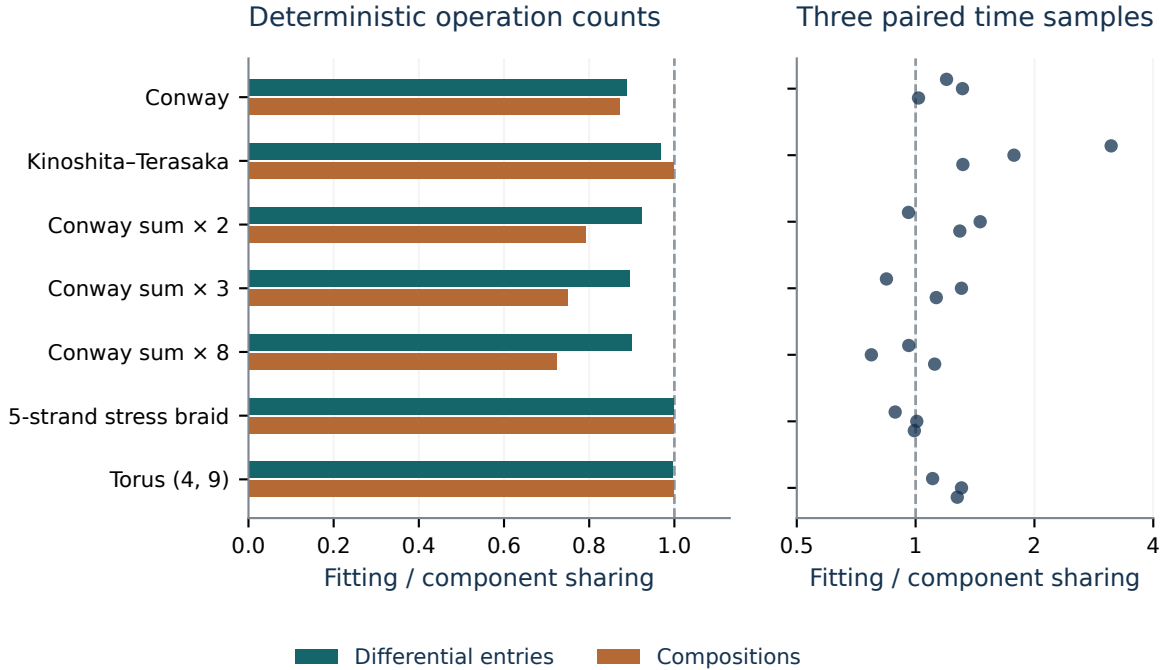


Figure 1: Structural work and wall time measure different effects. The left panel gives exact operation-count ratios for selected cases. Each right-panel dot is the Fitting/component time ratio in one interleaved repetition on that case; the time axis is logarithmic. Values below one favor Fitting. The three dots are measurements, not a confidence interval.

9.4 Wall time, the lazy implementation, and negative results

The final implementation performs no component copy before an accepted Fitting split. It does not rebuild the complete arrays if the Fitting or interval pass makes no change, and it reuses the verified component inventory across the passes. This matters on diagrams where the search is inactive. The preceding implementation copied skipped components and exhibited a large stress-case regression. The archived before/after audit compares 76 completed result summaries and 380 deterministic statistics with no differences. Both timing rounds are retained; only the final round supplies the main tables.

In the final round, exact Fitting takes a median 131.92 ms on the eight-fold Conway sum versus 138.30 ms for component sharing, an observed reduction of about 4.6%. The three Fitting samples span 114.07–132.88 ms, while the component samples span 102.08–170.76 ms. The overlap is substantial. This small run does not establish a stable speedup. Single Conway and the three-fold sum are slower with Fitting, at 12.03 versus 10.04 ms and 47.89 versus 42.45 ms, respectively. The stress case is nearly unchanged, 613.22 versus 615.47 ms. The implementation has removed an

avoidable overhead, but has not demonstrated a general wall-clock improvement over the preceding component backend.

Table 2: Median raw scanner time in milliseconds. S : standard; C_s : preceding component sharing; F : exact Fitting; F_3 : capped decision Fitting. DP preprocessing is included in the last column. lim denotes three resource-limited calls.

Case	n	S	C_s	F	F_3	$F_3 + \text{DP}$
U_8	8	2.14	1.35	2.37	1.45	2.28
C	11	8.94	10.04	12.03	12.53	17.18
KT	11	10.16	8.21	12.44	14.51	17.91
$C^{\#2}$	22	108.70	23.50	29.36	30.83	38.19
$C^{\#3}$	33	lim	42.45	47.89	46.52	61.84
$C^{\#8}$	88	lim	138.30	131.92	146.79	198.38
$B_{5,36}$	36	520.16	615.47	613.22	635.70	867.14
$T(3, 11)$	22	12.02	12.82	12.80	9.84	17.73
$T(4, 9)$	27	50.35	54.62	71.38	63.87	70.00
$T(5, 8)$	32	545.94	618.38	607.75	629.15	619.70
R_1	21	8.01	10.14	13.06	12.24	21.30
R_2	23	22.25	25.00	26.48	27.92	38.11
R_3	16	2.85	3.00	4.45	3.26	8.85
R_4	24	17.85	24.01	25.33	29.69	31.77
R_5	23	13.78	12.54	18.14	16.07	27.06
R_6	14	2.30	2.48	2.98	4.48	6.81
R_7	20	6.37	7.64	10.14	10.85	14.86
R_8	24	25.25	31.51	35.98	37.36	49.77
R_{24}	20	9.68	12.62	16.65	16.60	31.43

In Table 2, U_8 denotes the hard unknot, C Conway, KT Kinoshita–Terasaka, $C^{\#r}$ the supplied r -fold connected-sum diagram, and $B_{5,36}$ the supplied stress braid. The random labels are the valid-generation indices. Millisecond values below a few milliseconds are particularly sensitive to interpreter and host variation. The archive contains every sample, rather than only a best-of-three value.

Decision Fitting also has mixed timings. On this corpus, its difference from the exact variant concerns multiplicity saturation and degree bookkeeping; the new interval-length quotient never activates on a nontrivial actual component. The standard scanner remains the default, while the two new backends are explicit research options.

9.5 The interval quotient is currently demonstrated on synthetic complexes

The most important negative structural observation is exact: across all nineteen actual diagrams, the interval pass recognizes zero nontrivial whole common-nilpotent components, including after the accepted scalar splits. There is consequently no measured knot-diagram speedup attributable to the new length-two theorem. Actual residual towers carry additional matching types or differential attachments, so the whole-component hypothesis is restrictive. Extending the valid continuation quotient to those attached structures is a principal next research problem.

To verify the implemented transformation itself, the experiment also uses explicitly constructed legal cobordism complexes. In the first family, eight layers each contain q copies of one matching, with invertible bidiagonal scalar matrices multiplied by one dot endomorphism. The differential

graph is connected before the basis change. Exact interval normalization and sharing turn $8q$ stored objects into eight; decision shortening reduces those eight to two. At $q = 128$, the observed counts are therefore $1024 \rightarrow 8 \rightarrow 2$.

The second constructed family mixes two matching types and hides q copies of a three-object complex by a triangular scalar change of basis. For $q = 16$, certified Fitting splitting and exact sharing reduce 48 stored objects to three. This family tests a transformation that pure-component recognition cannot perform. Its normalization takes about 83 ms at that point, illustrating again that a dramatic representation reduction does not by itself establish an economical search procedure.

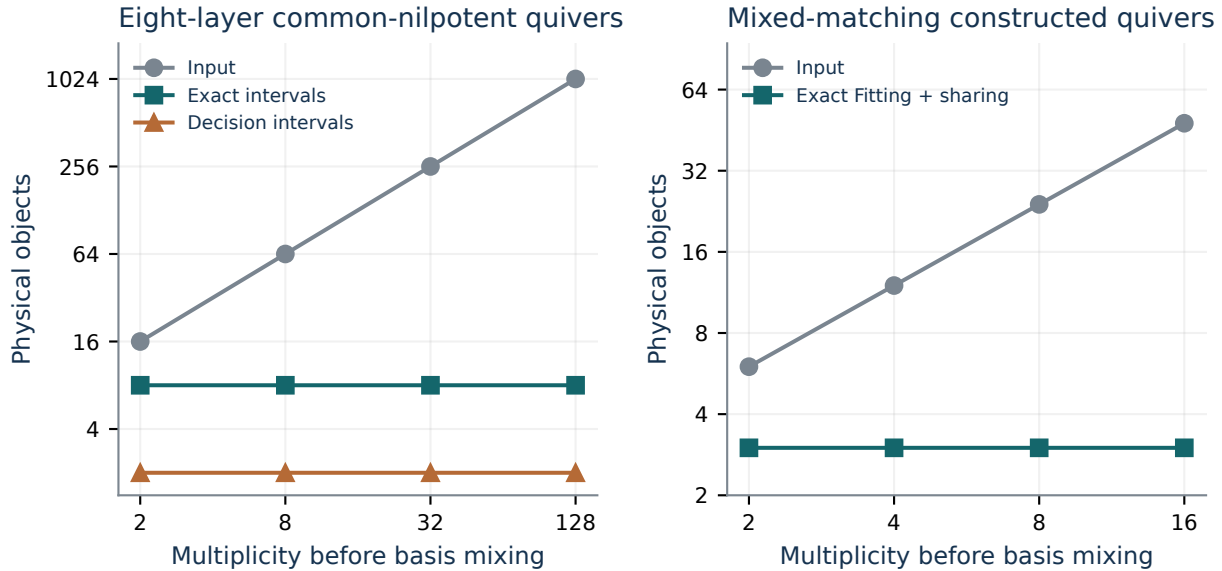


Figure 2: Physical object counts in two constructed algebraic families. Both axes are logarithmic. These are valid test complexes in the scanner’s category; their realization as intermediate complexes of actual knot diagrams is not asserted.

9.6 Checkpoint and ordering diagnostics

The decision scanner records a checkpoint only if every residual component has actually been classified and normalized as required by the theorem. The initial state is counted; the final closed state is also a checkpoint. On single Conway, the maximum observed gap is eleven crossings; the connected sums retain gap eleven as their number of summands grows. The stress braid has gap 36, while $T(3, 11)$, $T(4, 9)$, and $T(5, 8)$ have gaps 22, 27, and 32. Their respective maximum frontiers are 8, 6, 8, and 10. These small measurements support neither a universal logarithmic gap nor an extrapolation to all hard prime diagrams.

Window optimization improves the declared frontier-mass objective on eight of the nineteen cases. The peak frontier does not change in these cases. For example, the Kinoshita–Terasaka mass decreases from 61 to 57, and the eight-fold Conway sum decreases from 575 to 519. Every accepted order is verified and all resulting completed ranks agree. However, the preprocessing cost usually exceeds any downstream saving for a single scan. The stress case even becomes slower after its mass decreases, showing that the proxy does not capture the full differential structure.

The empirical conclusion is therefore specific. Certified scalar splitting reduces actual algebraic work on some diagrams. Lazy execution reduces its avoidable overhead. Exact window optimization

Case	Initial mass	Final mass	DP time (ms)	Full time (ms)
KT	61	57	7.52	17.91
$C^{\#2}$	131	123	15.29	38.19
$C^{\#3}$	209	189	24.95	61.84
$C^{\#8}$	575	519	86.86	198.38
$B_{5,36}$	369	353	28.22	867.14
R_2	225	221	14.44	38.11
R_8	165	161	15.36	49.77
R_{24}	141	125	11.32	31.43

Table 3: Every measured case whose frontier-mass objective strictly improves under window DP. Full time includes DP and the decision scan.

improves its own objective. The interval continuation quotient has a proved asymptotic benefit on its stated class and is correctly exercised by constructed complexes. A larger class of active normal forms and a structural checkpoint bound remain necessary before these ingredients justify a general complexity improvement.

10 Further research questions and a staged programme

The general target remains open for this implementation. The results above suggest concrete next questions that can be tested or proved without assuming the desired state bound. The order below reflects their dependence on the present code and mathematics.

10.1 Find the summands before they become large

The Fitting pass already detects summands that graph connectivity misses. Its cost currently competes with the time saved by sparser later matrices. Can the scalar endomorphism space be maintained incrementally when a crossing is added, instead of solved anew? A useful result would give a complete update algorithm with cost polynomial in the changed blocks, together with a criterion deciding when reusing an old splitting remains valid. Every reused projection must still commute with the new differential.

The current bounded candidate search is deliberately incomplete. A candidate endomorphism may be invertible or nilpotent on an indecomposable summand and give no split, even if another endomorphism would work. A next implementation could use primary decomposition of endomorphisms, or a proved decomposition algorithm for finite-dimensional module representations. Its cost must be compared with continuing the unsplit complex. The practical success criterion is a reduction in complete measured work including decomposition overhead, not the mere production of additional summands.

10.2 Extend the class with a finite continuation observation

The common- θ condition reduces the problem to the dual numbers. What is the next useful class? Two independently active dot directions lead toward

$$\mathbb{F}_2[x, y]/(x^2, y^2),$$

where the classification problem is substantially different. A carefully scoped goal is to classify continuation observations of a specified family of two-direction complexes at the rank-two threshold,

rather than classify all modules. The correct statement must quantify over every allowed suffix. Agreement on sampled closures is evidence for a conjecture, not a valid replacement rule.

Another useful target is a small tangle with two or more matching types and a restricted set of saddle maps. The attached towers seen in actual scans motivate this direction: whole-component common- θ recognition is too restrictive on the measured corpus. A normal form for this larger class could make the present quotient active on ordinary diagrams. Rational-tangle and four-ended tangle constructions provide relevant prior structure [17, 18]; their hypotheses and grading conventions must be retained.

10.3 Prove a structural bound on the unresolved intervals of a scan

The checkpoint theorem reduces the target to explicit bounds on w and g . An immediate mathematical question is whether a natural family of validated diagrams, extending the already tractable braid and rational-tangle families, admits a constructible order with

$$g + 2^{w/2} + w \log(w + 1) = O((\log n)^2).$$

A useful theorem must also bound the cost of finding that order. Merely asserting the existence of a favorable representation would not complete the algorithmic argument.

Corollary 5.4 provides another concrete target: logarithmic frontier width and $O(\log^2 n)$ checkpoint gaps suffice when the checkpoint coefficients have dot degree at most two. Can a marked, graded, or restricted-tangle structure force that degree bound after normalization? The condition concerns actual checkpoint coefficients; arbitrary intermediate algebra must still be charged in full.

A universal low-width replacement theorem for all knot types is unavailable: Lunel-de Mesmay give torus-knot families with unavoidable large diagram treewidth [15]. Such lower bounds do not prevent an algorithm from rejecting those knots using an inexpensive independent certificate. This suggests an asymmetric target: either produce a short nontriviality certificate, or construct a sequence of suitably bounded algebraic checkpoints. Establishing this dichotomy would be far stronger than optimizing a handful of observed scans.

10.4 Build a catalogue of sharp obstructions

The length-two bound has a sharp formal-block example. Similar sharpness examples are needed for every proposed observation quotient. In particular, one should construct common-nilpotent and mixed-matching families whose serialized component sizes grow, and determine which are realizable inside actual knot scans. A representation that is legal algebraically need not be the Khovanov complex of a tangle. Keeping these two questions separate will prevent synthetic compression demonstrations from being misreported as knot benchmarks.

The 2026 explicit-basis exponential examples [11] are especially valuable stress families. For each one, record how multiplicity, component size, scalar endomorphism dimension, and checkpoint gaps grow after each new normalization. The point is to identify a surviving obstruction, or prove that an observation quotient removes it, rather than infer asymptotics from a short timing curve.

10.5 Exploit graded or marked structure with compatible gluing

The present exact-rank backend retains homological degree but forgets quantum degree; decision compression discards still more information. A separate graded implementation could restrict scalar basis changes to compatible bigraded blocks and use the grading to constrain possible endomorphisms.

This may lower the cost of the Fitting solver, but could also remove useful ungraded decompositions. Both outcomes should be measured.

Cohen’s characteristic-two splitting of arc algebras is relevant to reducing the size of the coefficient algebra [16]. The tangle-bimodule result does not automatically preserve both actions in arbitrary configurations. A concrete next problem is to specify compatible marked endpoints throughout the scan and prove that the chosen identifications commute with every gluing operation. An algebra isomorphism alone does not supply that compatibility.

10.6 Add a complete tractable topological class

The linear-time recognition theorem for diagrams of treewidth two provides an independent complete structural backend [14]. Implementing its certified reductions would broaden the front end and offer a useful comparison with the algebraic route. A proper implementation should accept a verified decomposition, record every reduction, and distinguish recognition on the promised class from attempting to discover a suitable diagram.

10.7 Connect back to compressed hierarchy operations

The geometric route remains valuable. If an implementation has at most $L(g+1)^L$ phases and each phase costs at most B bit operations, then

$$\log_2 T \leq \log_2 B + \log_2 L + L \log_2(g+1) + O(1).$$

Here the letters are the hierarchy parameters, not the checkpoint gap of this article. This elementary accounting identity specifies what is needed from the hierarchy programme: controlled length, controlled pattern data, and bounded exact operations on the compressed encoding. The July 2026 preprint supplies important pieces [13]; it does not justify omitting any of these implementation costs.

10.8 Formalize the new algebraic interface

The continuation theorem is a manageable formalization target. It can be split into finite-dimensional linear algebra over $\mathbb{F}_2$, finite interval decomposition, the dual-number tensor pairing, and the parity refinement. The largest foundational dependency is the derived classification; the appendix provides an explicit elementary route that can be used instead of importing a general derived-category classification theorem.

A practical first verification milestone is an independently checkable certificate for each scalar basis split: binary change-of-basis matrices, their inverses, and the transformed differential. For interval normalization, rank certificates for the products in the multiplicity formula would certify the inferred decomposition. Formalizing the observation-preserving shortening rule then allows a small verifier to accept a decision model without trusting the optimizer that found it.

A An elementary dual-number derived classification

This appendix proves the classification used in Theorem 4.2, retaining the finiteness conditions needed for that application. The classification is established mathematics; see Keller–Yang–Zhou [8, Example 3.7(b)]. The argument here explains why the possible unbounded projective resolution introduces only simple-module summands, rather than additional unknown types.

Let $A = \mathbb{F}_2[\epsilon]/(\epsilon^2)$. A cochain complex has differential raising its degree by one. A finite free interval J_l has l free rank-one terms and consecutive differentials ϵ . A left-infinite interval ending in degree b is

$$\cdots \xrightarrow{\epsilon} A \xrightarrow{\epsilon} A \xrightarrow{\epsilon} A \xrightarrow{b} 0.$$

It is exact below b and has cohomology $A/(\epsilon) = \mathbb{F}_2$ in degree b . It therefore represents a shift of the simple module.

Proposition A.1 (Classification needed for continuation ranks). *Every bounded complex D of finite-dimensional A -modules is quasi-isomorphic to a finite direct sum of homological shifts of finite free intervals J_l and of the simple module $\mathbb{F}_2$.*

Proof. We give the construction in three steps.

Step 1: a degreewise finite free resolution. There is a bounded-above complex P of finite free A -modules and a quasi-isomorphism $f : P \rightarrow D$. The following downward construction establishes the precise finiteness assertion. Suppose P and f have been constructed in degrees greater than i . The degree- i term of the partially built mapping cone is $D^i \oplus P^{i+1}$, whose cycles are the finite A -module

$$Z_i = \{(d, p) : d_D d + f^{i+1} p = 0, \quad d_P^{i+1} p = 0\}.$$

Choose a finite free module P^i surjecting onto Z_i , and define f^i and d_P^i to be its two component maps. The defining equations of Z_i give the chain-map identity and $d_P^{i+1} d_P^i = 0$. The new boundaries from P^i in the cone cover all its degree- i cycles. Begin above the highest nonzero degree of D and continue downward. Each cone degree becomes exact, with its outgoing maps unaffected by later steps. Hence the completed cone is acyclic. The resulting P is bounded above and has finite free rank in every degree, as claimed.

Step 2: minimalization and the exact left tail. Over the local ring A , cancel every unit entry of the differential, working downward from the highest degrees. A unit pivot splits off a contractible pair $A \xrightarrow{\sim} A$. Each pair of adjacent degrees has finite rank, so only finitely many cancellations are needed there. Once a differential has all its entries in the radical (ϵ) , later changes of basis cannot make their residues nonzero. The descending procedure is locally finite: a fixed degree and its adjacent matrices stabilize after finitely many operations. It therefore gives a bounded-above minimal free complex P' quasi-isomorphic to P . This also follows by tracking the elementary cancellation equivalences degree by degree; no infinite sum of maps in a fixed degree is required.

Write $P'^i = A^{r_i}$. Every differential is now uniquely of the form

$$d_{P'}^i = \epsilon B_i, \quad B_i : \mathbb{F}_2^{r_i} \longrightarrow \mathbb{F}_2^{r_{i+1}}.$$

Multiplication by ϵB_i has binary rank $\text{rank}(B_i)$: it maps the constant part to the ϵ part and kills the latter. Consequently

$$\dim_{\mathbb{F}_2} H^i(P') = 2r_i - \text{rank}(B_{i-1}) - \text{rank}(B_i). \quad (21)$$

Because D is bounded, the left side is zero for every sufficiently negative i . Each rank on the right is at most r_i , so exactness in degree i forces

$$\text{rank}(B_{i-1}) = \text{rank}(B_i) = r_i.$$

At two consecutive exact degrees this gives $r_i = r_{i+1}$ and makes B_i invertible. Hence the sufficiently far-left tail has one constant finite rank r , with every B_i invertible. Choose bases recursively to make all these tail matrices identities. The case $r = 0$ simply means that the minimal complex is bounded on both sides.

Step 3: one finite interval decomposition. Choose a cut degree a within that identity tail. Only finitely many nonzero degrees lie at or to the right of a , and their dimensions are finite. Apply the elementary finite interval decomposition from Theorem 3.3 to the sequence of binary matrices from degree a to the highest degree of P' .

Every interval born strictly after a remains a finite interval. Every interval born at a extends to the left along one of the identity tail coordinates. To see this explicitly, extend the chosen change of basis in degree a backward through the identity tail; all its matrices remain identities. Thus these intervals become left-infinite intervals, with no additional couplings. There are finitely many intervals altogether because the finite sequence being decomposed has finite total dimension.

Reinsert the factor ϵ in the differentials. This expresses P' as a finite direct sum of finite free ϵ -intervals and left-infinite ϵ -intervals. Each left-infinite interval is quasi-isomorphic to a shifted simple module, as observed above. Composing with the quasi-isomorphisms in the first two steps proves the proposition. $\square$

Only existence is needed in the continuation argument. The recognizer does not construct this resolution for a suffix or attempt to process an infinite complex. It computes the finite barcode of an already stored pure component, then applies the rank identity which the classification proves for every possible suffix.

The distinction between free and arbitrary module continuations matters. If D is perfect, only finite free intervals occur, so the coefficient a in (8) vanishes. In an arbitrary bounded module complex, shifted simple modules can occur. The geometric cutoff-two theorem did not assume their absence: it proved that their number a is even, using the unreduced circle-space parity of the actual closure. This is why that theorem applies to every square-zero dot polynomial, including ones whose action is not free on a resolution.

B Reproducibility and artifact boundaries

The archive contains the complete `fast/` source tree, this article's $\text{\LaTeX}$ sources and PDF, deterministic verification programs, paired timing data, two integration patches, and source provenance. The main package `README.md` gives the shortest entry points; `REPRODUCING.md` describes each experiment and its scope.

The software uses the Python standard library. From `fast/`, run

```
python -m unittest discover -s tests -v
python -m fastunknot recognize examples/conway.json --backend fitting
python -m fastunknot khovanov examples/conway.json --fitting
```

The last command computes exact ranks with full interval lengths. The recognition command may return through a preceding structural or polynomial filter. To reproduce raw backend measurements, use the benchmark script, whose inputs, backend options, and order selection are recorded in its JSON.

From the package root, the independent continuation check is

```
python verification/verify_continuation.py
```

It imports no recognizer module. It enumerates small bounded complexes with strict square-zero actions, constructs tensor differentials directly as binary matrices, checks their squares, and compares homology thresholds for different interval lengths. It also records the counterexamples to an incorrect universal length-two cutoff without parity and to a geometric length-one cutoff.

The article can be rebuilt using

```
sh scripts/build_pdf.sh
```

This requires a standard L^AT_EX installation with the packages listed in the preamble. All figure PDFs and generated table fragments needed by the article are included. Plot regeneration additionally uses Matplotlib; it is not required to run the recognizer or its mathematical checks.

B.1 What the records establish

The source manifest establishes exact byte identity with the stated upstream files. The patch verification establishes that the supplied textual deltas reproduce the delivered source from those references. Unit tests establish the stated finite comparisons; paired timings establish observations on the recorded host and inputs. None is a substitute for the mathematical continuation theorem, a proof that every diagram satisfies the checkpoint condition, or a claim of external peer review.

The included witness data certify individual basis changes, including all incident maps. An emitted witness is local evidence for a split. The final recognizer still relies on the established Khovanov unknot-detection theorem and the implementation of the inherited cobordism algebra.

B.2 Integration order

Use the cumulative patch for the pinned repository tree, or the incremental patch for the already integrated structural 0.3.0 tree. Do not apply both. The patches target `Topology/UnknotRecognition/fast/`. Changes in a different branch’s front-end files should be merged before running the full combined suite. All reference material redistributed in the package is project source under the supplied MIT-0 license; third-party papers and lecture slides are referenced by bibliographic links.

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